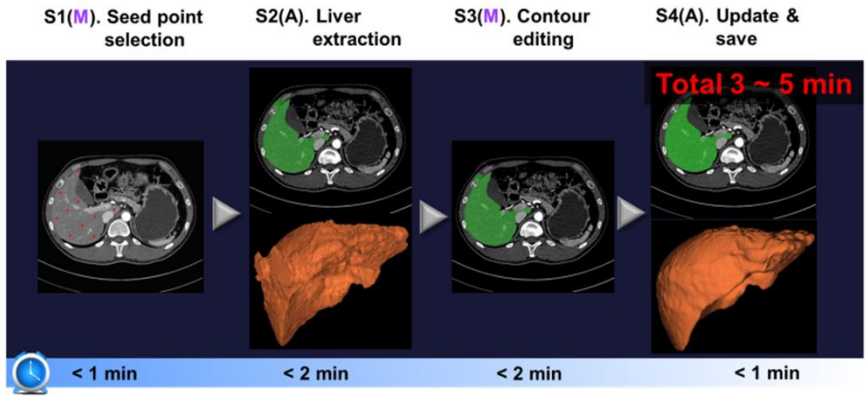
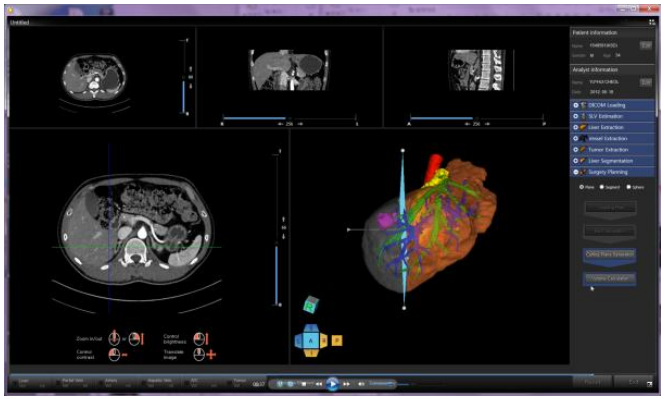
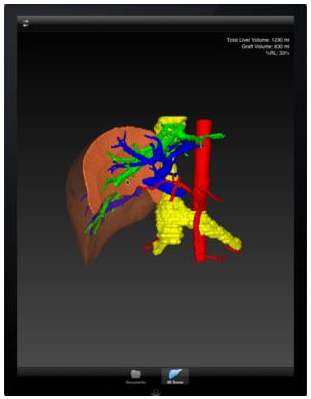
Information Input & Processing







Virtual liver surgery planning system (You et al., Jan. 2009 ~ present; sponsored by Ministry of Health & Welfare & Humanopia Co., Ltd.)

Contents

Part I

- Information Theory
- Information Display & Modality Selection
- Information Coding & Judgment
- Signal Detection Theory (SDT)

Part II

- Human Memory Systems
- Attention & Vigilance
- Compatibility & Aging
- Mental Workload & AI
- Quiz

Learning Goals (1/2)

GOALS

- Define **information** as the **reduction of uncertainty** and explain **the unit of measure (bit)**.
- Compute **the amount of information (*H*)** in bits for alternatives with designated probabilities.
- Apply **the Hick-Hyman Law** to predict **choice reaction time** in system design.
- Distinguish between **Static and Dynamic displays** and identify **various information types**, including Quantitative, Qualitative, and Status (Check-Reading).
- Evaluate the trade-offs between **visual and auditory displays** based on message complexity, urgency, and environmental conditions.
- Implement **multimodal safety strategies**, prioritizing auditory signals for initial alerting and visual displays for detailed diagnosis.

- Differentiate between **Absolute Judgment and Relative Judgment**.
- Apply **Miller's Law (7 ± 2)** to limit **the number of levels in a single coding dimension** to prevent identification errors.

Learning Goals (2/2)

GOALS

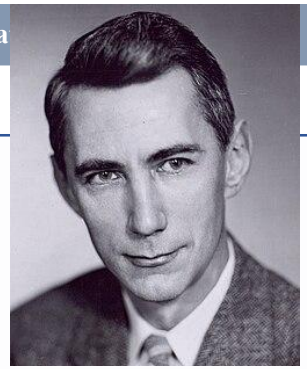
- Compare **Orthogonal Coding** for maximum data density with **Redundant Coding** for increased detectability and safety.
- Define and apply the **four SDT outcomes—Hit, False Alarm, Miss, and Correct Rejection**—to diagnostic scenarios.
- Calculate the **Sensitivity (d')** to measure discriminability and evaluate the **Response Criterion (β)** to understand an observer's decision strategy.
- Use **ROC Curves and outcome matrices** to calibrate decision thresholds in high-stakes environments by considering the **"costs and benefits" of outcomes**.

Section 01

Information Theory & Quantifying Information

From Uncertainty to Bits • Hick-Hyman Law

What is Information?



Claude E. Shannon
(1916 ~ 2001)
"father of information theory"
Mathematician, electrical engineer, computer scientist, and cryptographer

Key Definition

Information is defined as the **reduction of uncertainty**. It quantifies how much "surprise" a message conveys based on probability.

Probability vs. Information

Highly Likely Events: Convey *little* information because they confirm what was expected.

Example: "The sun rose today." (Probability ≈ 1.0)

Highly Unlikely Events: Convey *more* information because they significantly reduce uncertainty.

Example: "It snowed in the Sahara." (Probability ≈ 0.0)

Information Quantity ≠ Importance

Information theory **measures quantity (bits)**, **not semantic meaning or value**.

A message can have high importance but low information quantity, and vice versa.

"Fire Alarm!"
High Importance
High Information (Rare)

VS

"It is 12:00 PM"
Low Importance
Low Information (Expected)

Measuring Information

Basic Information Formula

$$H = \log_2 N$$

H = Amount of information (in bits)

N = Number of equally likely alternatives

$\log_2$ = Logarithm to the base 2

What is a Bit?

One bit is the amount of information required to decide between **two equally likely alternatives** (e.g., a coin flip).

Calculation Examples

Number of Alternatives (N)	Calculation ($\log_2 N$)	Information (H)
2 (Binary)	$\log_2(2) = 1$	**1 Bit**
4 (Quaternary)	$\log_2(4) = 2$	**2 Bits**
8 (Octal)	$\log_2(8) = 3$	**3 Bits**
16 (Hex)	$\log_2(16) = 4$	**4 Bits**

Key Insight: Every time the number of alternatives doubles, the information content increases by exactly **1 bit**.

Unequal Probabilities (Entropy)

Single Event Information

$$h_i = \log_2 \frac{1}{p_i}$$

h_i = Information of specific event i (bits)

p_i = Probability of occurrence for event i

Average Information (Entropy)

$$H_{av} = \sum_{i=1}^{N} p_i \left(\log_2 \frac{1}{p_i} \right)$$

H_{av} = Average information (Entropy)

$\sum_{i=1}^{N}$ = all possible alternatives

Calculation Example

Event (Probability)	Info ($\log_2 1/p_i$)	Contribution (p_i × Info)
A $p_A = 0.8$	$\log_2(1.25) \approx 0.32$	$0.8 \times 0.32 = 0.256$
B $p_B = 0.2$	$\log_2(5.00) \approx 2.32$	$0.2 \times 2.32 = 0.464$
Total (H_{av})	-	**≈ 0.72 Bits**

Why is $H_{av} < 1$ bit?

If A and B were equally likely (0.5 each), H_{av} would be **1.0 bit**.
Since Event A is highly probable (0.8), there is **less uncertainty** overall, resulting in lower average information (0.72 bits).

Redundancy

Redundancy Formula

$$\% \text{ Redundancy} = \left(1 - \frac{H_{av}}{H_{max}}\right)$$

H_{av} = Average information (Entropy) calculated from actual probabilities

H_{max} = Maximum possible information ($\log_2 N$) assuming equal probabilities

Ratio = H_{av}/H_{max} represents the relative efficiency

Concept Insight

Redundancy measures how much information is "lost" due to unequal probabilities. Higher redundancy means the message is more predictable.

Calculation Example

Given a system with **4 alternatives (N=4)** where probabilities are unequal such that average information is lower than maximum.

1. Maximum Information (H_{max})	**$\log_2(4)$ = 2.0 bits**
2. Average Information (H_{av})	**Given as 1.2 bits**
3. Efficiency Ratio (H_{av}/H_{max})	**1.2 / 2.0 = 0.6**

Calculated Redundancy

[1 - 0.6] × 100 = 40%

Design Implication: 40% redundancy means the system is only 60% efficient in carrying information, but this redundancy can help reduce errors in noisy environments.

Hick-Hyman Law

Choice Reaction Time

Reaction time increases linearly with the amount of information (bits) transmitted.

$$RT = a + b(H)$$

Reaction Time Formula

$$RT = a + b(H)$$

a = Intercept (Non-decision time)
(Sum of encoding time and movement execution time)

b = Slope (Processing speed)
(Time required to process 1 bit of information;
inverse of bandwidth)

H = Information (in bits)

Design Implications

The relationship is logarithmic with respect to the number of alternatives (N). **Doubling the number of alternatives adds a constant amount of time (*b*).**

Doubling N = +1 Bit = +*b* ms

Alts (N)		Alts (N)	Effect
4	→	8	+ *b*
H = 2		H = 3	Time

Example: Menu Selection Time

Problem Scenario

Calculate the estimated Reaction Time (RT) for selecting an item from a computer menu. Assume a simple reaction time intercept (***a* = 250 ms**) and a processing speed (***b* = 200 ms/bit**). Compare 4, 8, and 16 options.

Calculation Steps: RT = $a + b(H)$

Options (N)	Information ($H = \log_2 N$)	Calculation (250 + 200 × H)	Result (RT)
4 options	2 bits	250 + (200 × 2)	**650 ms**
8 options	3 bits	250 + (200 × 3)	**850 ms**
16 options	4 bits	250 + (200 × 4)	**1,050 ms**

Visual Breakdown (16 Items) Total: 1,050 ms

Movement/Motor Time Decision Processing Time

Design Implications

- **Processing Cost:** Every doubling of menu items adds a constant time cost (200ms in this case).
- **Broad vs. Deep:** Broad menus (many items) are generally faster than deep hierarchies because users scan faster than they navigate multiple screens.
- **Optimization:** Group items logically to reduce the effective search set size.

Design Implications

Applying Information Theory

1 Minimize Alternatives

In time-critical tasks, reducing the number of choices directly reduces decision time (RT).

- Remove non-essential options
- Context-sensitive menus

✓ **Less choice = Faster reaction**

2 Menu Structure

Balance breadth vs. depth. Broad menus are generally faster than deep hierarchies.

- Use "Mega Menus" over deep nesting
- Group logically (chunking)

✓ **Broad > Deep for speed**

3 Frequency & Defaults

Leverage probability to reduce average information load.

- Place frequent items at top
- Use smart defaults (predictive)

✓ **Optimize for common case**

Section 02

Information Display & Modality Selection

From dynamic vs. static to optimal sensory channels

Static vs. Dynamic Displays

Static Display

Definition

Information content remains **fixed** over time (or changes very infrequently).

- Road Signs / Labels
- Printed Charts / Graphs
- Alphanumeric Data (Text)

VS

Dynamic Display

Definition

Information content **changes continuously** or varies over time based on system state.

- Speedometer / Tachometer
- Radar Screen / GPS Navigation
- Traffic Signals / Status Lights

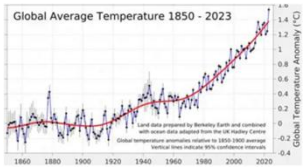
Types of Displayed Information

Quantitative

Numerical values or magnitude of some variable (e.g., temperature, speed, altitude).

Qualitative

Approximate value, trend, or direction of change (e.g., rising/falling, hot/cold).



Status (Check-Reading)

Condition or mode of a system (e.g., On/Off, Auto/Manual, Gear selection).

Warning/Signal

Alerting to unsafe conditions or emergency situations (e.g., Fire alarm, Low battery).

Representational

Pictorial or graphic representation of objects or areas (e.g., Maps, mimic diagrams).

Identification

Coded information to identify specific situations or objects (e.g., Traffic lane types).

Alphanumeric

Verbal or numerical information in text form (e.g., Labels, instructions, signs).

Time-phased

Signals controlled by time duration or sequence (e.g., Morse code, traffic light cycles).

Auditory vs. Visual Modality

Select the sensory channel based on message characteristics, task requirements, and environmental constraints.

Use Auditory Presentation	Use Visual Presentation
Message is **simple** and **short**.	Message is **complex** and **long**.
Message will **not be referred to later**.	Message will be **referred to later**.
Message deals with **events in time**.	Message deals with **location in space**.
Message calls for **immediate action**.	Message does **not** call **for immediate action**.
Visual system is **overloaded**.	**Auditory** system is **overloaded**.
The location is **too bright or dark**-adaptation integrity is necessary.	The location is **too noisy**.
The receiver's job requires **moving about**.	The receiver's job allows them to remain in **one position**.

Warning System Design

Critical Design Guidelines

1 Alert vs. Diagnose

Prioritize modalities based on their inherent strengths in the warning sequence.

- **Auditory:** Best for initial alerting (omnidirectional)
- **Visual:** Best for diagnosis & guidance (spatial/detailed)

✓ Hear the alert, See the problem

2 Auditory Specs

Ensure the signal cuts through noise and conveys urgency without startling.

- **Intensity:** ≥15 dB above ambient noise
- **Temporal:** 3-5 pulses/sec for high urgency
- **Frequency:** Distinctive timbre, avoid masking

✓ Detectable & Urgent

3 Visual Specs

Visual warnings must be conspicuous within the operator's field of view.

- **Location:** Within 30° of normal line of sight
- **Salience:** Use Red/Amber color + Flashing (3-5 Hz)
- **Size:** Subtends at least 1° visual angle

✓ Conspicuous & Meaningful

4 Multimodal Safety

For safety-critical hazards, do not rely on a single sensory channel.

- **Redundancy:** Present simultaneously via Audio + Visual
- **Reliability:** Ensures detection if one channel is blocked/overloaded

✓ Two senses are safer than one

Case Study: Glass Cockpit Alerts

The Challenge

Modern aircraft systems generate hundreds of potential alerts. In high-stress situations, simultaneous alarms can overwhelm the pilot's auditory and visual channels, leading to confusion or ignored warnings.

Design Solutions

- **Priority Tiers & Coding**
  Categorize alerts into Warning (Red/Siren), Caution (Amber/Chime), and Advisory (Cyan/Text) to guide attention.
- **Master Caution/Warning System**
  Use a central "Master" light/tone to alert the pilot to look at the specific diagnostic display (Annunciator Panel).
- **Inhibit Logic (Smart Alerts)**
  Suppress lower-priority alerts during critical flight phases (e.g., takeoff) to prevent distraction.

Cockpit Display

EICAS

Engine Indication and Crew Alerting System

WARNING
Immediate Action Required (Fire, Stall)

CAUTION
Immediate Awareness Required (Low Fuel)

ADVISORY
Information Only (Parking Brake Set)

Design Goal
Prevent "Cry Wolf" effect & reduce cognitive load.

Section 03

Information Coding & Absolute Judgment

Choosing stimulus dimensions to maximize detectability and meaning

What is Coding?

Definition: Information Coding

Coding is the process of converting original stimulus information into a new form by mapping raw data onto specific **display dimensions**. This symbolic representation allows operators to identify or distinguish items efficiently.

Common Coding Dimensions

Information is coded along various sensory dimensions. The choice depends on task requirements and human limits.

- Color (Hue)
- Shape (Form)
- Size (Area)
- Brightness
- Pitch / Loudness
- Text / Labels

Principles for Effective Coding

Detectability & Salience
Coding dimensions must be distinct enough to be noticed against the background (e.g., adequate contrast, size).

Discriminability:
Users must be able to tell code steps apart (e.g., distinguishing between yellow and amber).

Consistency & Standards:
Adhere to population stereotypes (e.g., Red = Danger) and maintain meaning across the system.

Absolute vs. Relative Judgment

Absolute Judgment

Definition

Identifying a stimulus based on its position along a sensory dimension **from memory**.

Limited Capacity: Humans are poor at this. Can only reliably identify 5-9 distinct levels (7±2).

Example: Identifying the exact pitch of a tone without a reference tone (Perfect Pitch).

VS

Relative Judgment

Definition

Comparing two or more stimuli presented **simultaneously** or in close succession.

High Precision: Humans are excellent at detecting small differences when a reference is present.

Example: Telling which of two colors is brighter when seen side-by-side.

Design Implication: Use Reference Cues

Whenever possible, convert absolute judgment tasks into relative judgment tasks by providing reference scales, comparison standards, or legends to improve accuracy and reduce cognitive load.

The Magical Number 7±2

Miller's Law (1956)

There is a fundamental limit to the number of items humans can hold in working memory or identify on an absolute basis. This span of absolute judgment is typically **7 items, plus or minus 2**.

Channel Capacity Limitations

Absolute Identification: The ability to label a stimulus (e.g., a specific pitch or color) without a comparison reference is poor.

Capacity in Bits: The "channel capacity" for unidimensional stimuli is approximately **2 to 3 bits**.

($\log_2 5 \approx 2.3$ bits to $\log_2 9 \approx 3.2$ bits)

Beyond the Limit: As the number of alternatives exceeds ~7, errors in identification increase sharply.

7 ± 2

5 (Safe)	7 (Avg)	9 (Risky)

Design Rule

To ensure error-free performance, limit distinct levels in a single code dimension to **5–7 items**.

[Paper] The Magical Number 7±2

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The Magical Number Seven, Plus or Minus Two
Some Limits on Our Capacity for Processing Information

George A. Miller
Harvard University

This paper was first read as an Invited Address before the Eastern Psychological Association in Philadelphia on April 15, 1955. Preparation of the paper was supported by the Harvard Psycho-Acoustic Laboratory under Contract N5ori-76 between Harvard University and the Office of Naval Research, U.S. Navy (Project NR 142-201, Report PNR-174). Reproduction for any purpose of the U.S. Government is permitted.
Received: May 4, 1955

My problem is that I have been persecuted by an integer. For seven years this number has followed me around, has intruded in my most private data, and has assaulted me from the pages of our most public journals. This number assumes a variety of disguises, being sometimes a little larger and sometimes a little smaller than usual, but never changing so much as to be unrecognizable. The persistence with which this number plagues me is far more than a random accident. There is, to quote a famous senator, a design behind it, some pattern governing its appearances. Either there really is something unusual about the number or else I am suffering from delusions of persecution.

I shall begin my case history by telling you about some experiments that tested how accurately people can assign numbers to the magnitudes of various aspects of a stimulus. In the traditional language of psychology these would be called experiments in absolute judgment. Historical accident, however, has decreed that they should have another name. We now call them experiments on the capacity of people to transmit information. Since these experiments would not have been done without the appearance of information theory on the psychological scene, and since the results are analyzed in terms of the concepts of information theory, I shall have to preface my discussion with a few remarks about this theory.

Information Measurement

The "amount of information" is exactly the same concept that we have talked about for years under the name of "variance." The equations are different, but if we hold tight to the idea that anything that increases the variance also increases the amount of information we cannot go far astray.

The advantages of this new way of talking about variance are simple enough. Variance is always stated in terms of the unit of measurement–inches, pounds, volts, etc.–whereas the amount of information is a dimensionless quantity. Since the information in a discrete statistical distribution does not depend upon the unit of measurement, we can extend the concept to situations where we have no metric and we would not ordinarily think of using the variance. And it also enables us to compare results obtained in quite different experimental situations where it would be meaningless to compare variances based on different metrics. So there are some good reasons for adopting the newer concept.

Summary

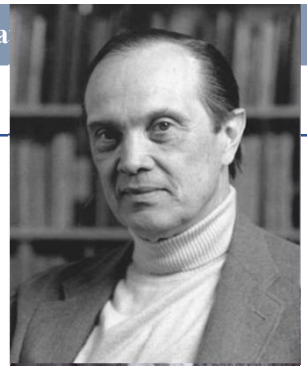
I have come to the end of the data that I wanted to present, so I would like now to make some summarizing remarks.

First, the span of absolute judgment and the span of immediate memory impose severe limitations on the amount of information that we are able to receive, process, and remember. By organizing the stimulus input simultaneously into several dimensions and successively into a sequence of chunks, we manage to break (or at least stretch) this informational bottleneck.

Second, the process of recoding is a very important one in human psychology and deserves much more explicit attention than it has received. In particular, the kind of linguistic recoding that people do seems to me to be the very lifeblood of the thought processes. Recoding procedures are a constant concern to clinicians, social psychologists, linguists, and anthropologists and yet, probably because recoding is less accessible to experimental manipulation than nonsense syllables or T mazes, the traditional experimental psychologist has contributed little or nothing to their analysis. Nevertheless, experimental techniques can be used, methods of recoding can be specified, behavioral indicants can be found. And I anticipate that we will find a very orderly set of relations describing what now seems an uncharted wilderness of individual differences.

Third, the concepts and measures provided by the theory of information provide a quantitative way of getting at some of these questions. The theory provides us with a yardstick for calibrating our stimulus materials and for measuring the performance of our subjects. In the interests of communication I have suppressed the technical details of information measurement and have tried to express the ideas in more familiar terms; I hope this paraphrase will not lead you to think they are not useful in research. Informational concepts have already proved valuable in the study of discrimination and of language; they promise a great deal in the study of learning and memory; and it has even been proposed that they can be useful in the study of concept formation. A lot of questions that seemed fruitless twenty or thirty years ago may now be worth another look. In fact, I feel that my story here must stop just as it begins to get really interesting.

And finally, what about the magical number seven? What about the seven wonders of the world, the seven seas, the seven deadly sins, the seven daughters of Atlas in the Pleiades, the seven ages of man, the seven levels of hell, the seven primary colors, the seven notes of the musical scale, and the seven days of the week? What about the seven-point rating scale, the seven categories for absolute judgment, the seven objects in the span of attention, and the seven digits in the span of immediate memory? For the present I propose to withhold judgment. Perhaps there is something deep and profound behind all these sevens, something just calling out for us to discover it. But I suspect that it is only a pernicious, Pythagorean coincidence.



George A. Miller
(1920–2012)
"founding father of cognitive psychology"
A faculty member at Harvard as well as MIT and Princeton

Absolute Judgment Capacities

Miller's "Magical Number 7±2" varies significantly across physical dimensions. Channel capacity is generally low for unidimensional stimuli.

Stimulus Dimension	Number of Levels	Information (Bits)
Pure Tones	5	2.3
Loudness	4 – 5	2.0 – 2.3
Size (Visual)	5 – 7	2.3 – 2.8
Brightness	3 – 5	1.7 – 2.3

Orthogonal vs. Redundant Coding

Orthogonal Coding

Definition

Dimensions are **independent** and used to encode **different variables** simultaneously.

- **Goal:** Maximize information capacity.
- **Example:** Shape = Aircraft Type, Color = Altitude.
- **Result:** Higher total identifiable states (N × M).

VS

Redundant Coding

Definition

Multiple dimensions are used to encode the **same variable** or message.

- **Goal:** Maximize detectability & reliability.
- **Example:** Red Color + Octagon Shape + "STOP" Text.
- **Result:** Reduced errors, robust against noise.

Strategic Selection Rule

Use **Orthogonal** for maximum data density (capacity). Use **Redundant** when safety, speed, or adverse conditions (noise, low light) are critical factors.

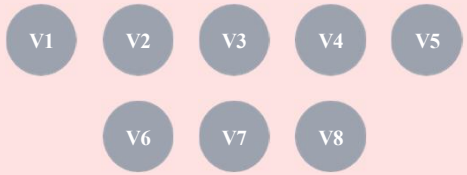
Example: Control Panel Coding

Design Challenge

Improve the discriminability of critical valves on a chemical plant control panel. The current baseline design uses 8 similar knobs distinguished only by small labels and slight color variations, leading to high selection error rates.

Visual Comparison

✕ Baseline (Poor)



Issues: Single dimension (position/label only), low discriminability, high cognitive load to identify.

✓ Improved (Robust)

Solution: Multi-dimensional coding (Shape + Color + Size + Texture).

Performance Result

Significantly reduced selection errors and faster access time due to redundant cues and categorical perception.

Orthogonal Coding

Mapping dimensions to functions

- **Shape = Function Group**
  (Square: Flow, Triangle: Pressure)
- **Color = Priority**
  (Blue: Normal, Red: Emergency)
- **Size = Control Level**
  (Large: Main, Small: Fine-tune)

Redundant Safety

For critical controls

- **Combination:** Color (Red) + Unique Shape + Raised Texture
- **Benefit:** Ensures identification even if one channel fails (e.g., in low light or for color-blind operators).
- **Tactile Feedback:** Texture allows "eyes-free" confirmation.

Design Rules

Coding & Absolute Judgment Guidelines

1 Single Dimension Limits

Respect human cognitive limitations for absolute judgment.

- Keep single-dimension categories ≤ 7 (Miller's 7±2)
- Use relative comparisons (scales) for finer discrimination

✓ **Respect 7±2 Limit**

2 Orthogonal Combination

To increase information capacity, combine independent dimensions.

- e.g., Shape × Color = More identifiable states
- Allows users to filter by either dimension

✓ **Maximize Capacity**

3 Redundancy & Safety

Use multiple coding dimensions for the same variable in critical tasks.

- Color + Shape + Text
- Ensures detection in noise or for users with CVD (Color Vision Deficiency)

✓ **Ensure Detectability**

Section 04

Signal Detection Theory (SDT)

Decisions under uncertainty: performance, bias, and payoffs

The Detection Problem

Core Challenge: Overlapping Responses

The world has two discrete states (**Signal** vs. **Noise**), but the human sensory system produces a continuous internal response. These responses often **overlap**, creating uncertainty in decision-making.

Two States of the World

Signal Absent (Noise Only): The background environment. "Noise" is always present in any system (neural, environmental, or equipment noise).

Signal Present (Signal + Noise): The target stimulus added to the background noise. The task is to detect this addition.

The Dilemma: Because noise is random, a strong noise event can look like a weak signal, leading to errors.

Real-World Scenarios

Medical Imaging

Tumor (Signal) vs.
Dense Tissue (Noise)

Security Screening

Weapon (Signal) vs.
Cluttered Bag (Noise)

Sonar / Radar

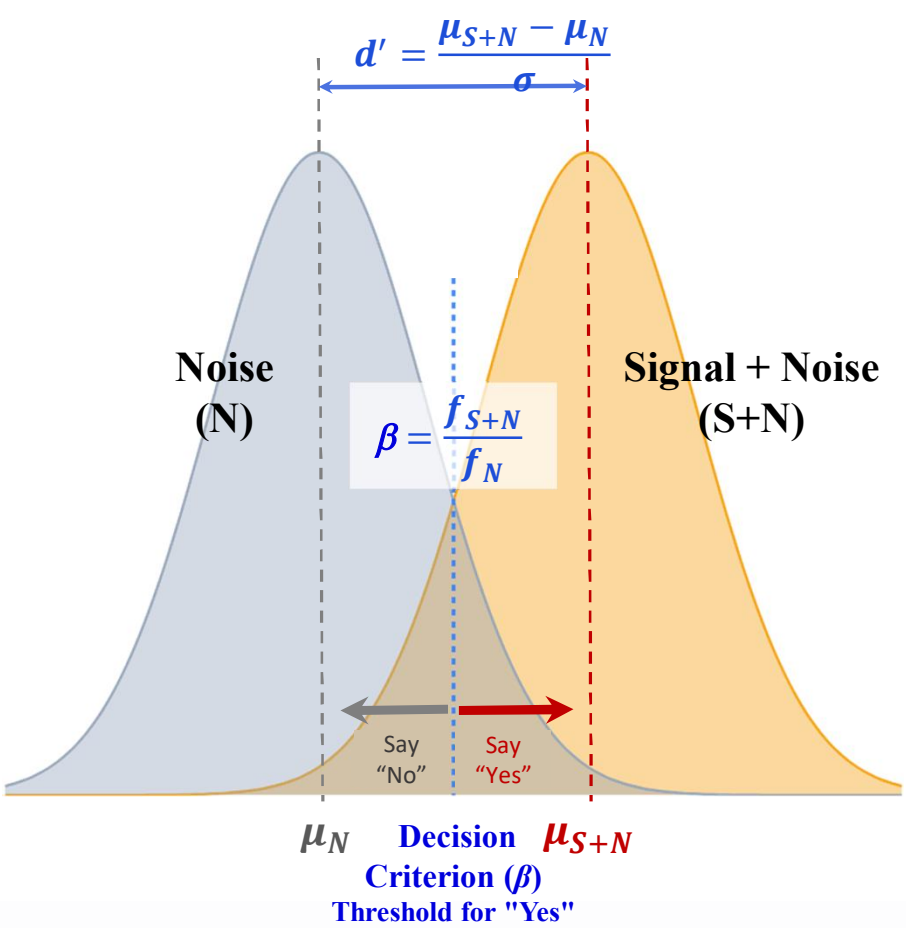
Enemy Sub (Signal) vs.
Marine Life (Noise)

Quality Control

Defect (Signal) vs.
Normal Variation (Noise)

SDT Framework

Internal Response Distributions



Noise (N)

Random background activity (neural or environmental) that is always present. The system must filter this out.

Signal + Noise (S+N)

The target event added to the background noise. Creates a shifted distribution of sensory evidence.

Decision Criterion (β)

The internal threshold set by the observer.
Right of line: Respond "Yes" (Signal)
Left of line: Respond "No" (Noise)

SDT Outcomes: 2×2 Matrix

Key Principle: In any detection task, these four outcomes are exhaustive. Increasing Hits often increases False Alarms (trade-off), while Sensitivity (d') measures the ability to maximize correct decisions (Hits + Correct Rejections).

State (Truth)	Response: Respond "YES" (Signal)	Response: Respond "NO" (Noise)
Signal Present	Correct Decision **HIT** Correctly detecting a signal when it is present.	Error Type II **MISS** Failing to detect a signal when it is present.
Signal Absent	Error Type I **FALSE ALARM** Incorrectly indicating a signal when none is present.	Correct Decision **CORRECT REJECTION** Correctly indicating that no signal is present.

Sensitivity (d')

Sensitivity Index Formula

$$d' = \frac{\mu_{S+N} - \mu_N}{\sigma}$$

d'	=	Sensitivity index (discriminability)
μ_{S+N}	=	Mean of Signal + Noise distribution
μ_N	=	Mean of Noise distribution
σ	=	Standard deviation (assumed equal)

**Assumes equal variance for both distributions*

Separability & Accuracy

The sensitivity index d' measures the distance between the signal and noise distributions in standard deviation units.

- **Larger d'** indicates better separability between signal and noise.
- **Higher d'** leads to **higher accuracy** regardless of the decision criterion (β).
- $d' = 0$ means the signal is indistinguishable from noise (chance performance).

Typical Values

In most applied settings, d' values typically range from **0.5 to 2.0**.

0 (Chance) 1.0 (Moderate) 3.0+ (Perfect)

Typical Applied Range (0.5 - 2.0)

Improve d' by: Increasing signal strength, reducing noise, enhancing display quality, or training operators to better recognize signal features.

Response Criterion (β)

Decision Bias

Beta (β) quantifies the observer's decision strategy (bias). It is the ratio of the neural activity height of the *Signal+Noise* distribution to the *Noise* distribution at the chosen criterion cut-off.

Conservative Strategy (β > 1)

The operator sets a high threshold and says "Yes" only when absolutely certain. Common when False Alarms are costly.

"Better safe than sorry (don't cry wolf)."

False Alarms	↓ **Fewer (Good)**
Misses	↑ **More (Bad)**
Correct Rejections	↑ **More**

Liberal / Risky Strategy (β < 1)

The operator sets a low threshold and says "Yes" at the slightest hint of a signal. Common when Misses are dangerous.

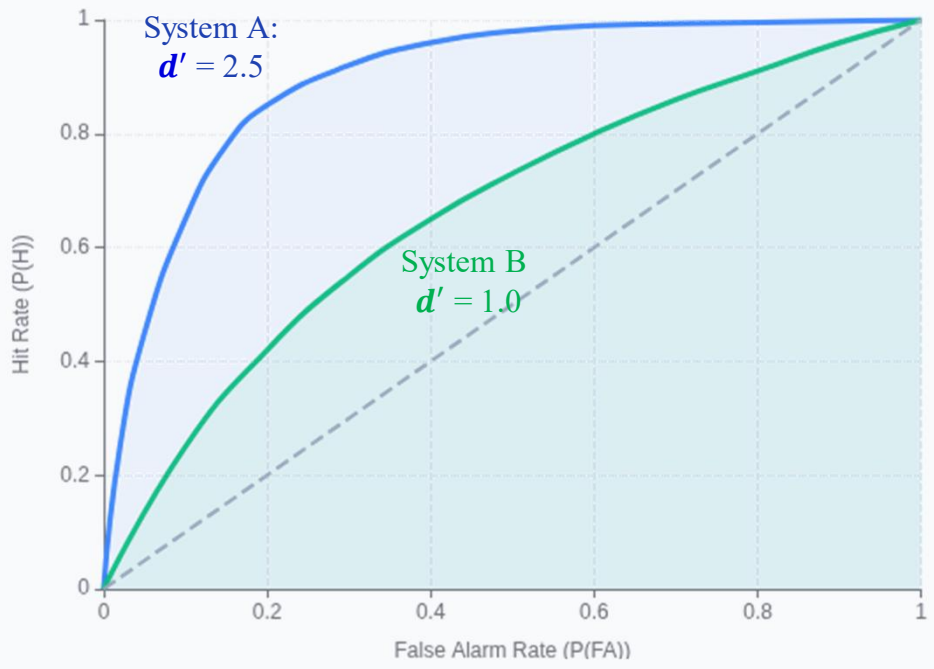
"Better safe than sorry (don't miss it)."

Hits	↑ **More (Good)**
False Alarms	↑ **More (Bad)**
Correct Rejections	↓ **Fewer**

Factors Shifting β

Payoffs (Costs & Benefits) Signal Probability (Prevalence)

ROC Curves & AUC



Receiver Operating Characteristic (ROC)

A graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied.

- ↑ Y-Axis:True Positive Rate (Hit Rate)
- → X-Axis:False Positive Rate (False Alarm Rate)

Area Under Curve (AUC)

A single scalar value representing expected performance. **AUC = 0.5 suggests random guessing, while AUC = 1.0 represents a perfect classifier**.

System A (Blue): $d' = 2.5$ System B (Green): $d' = 1.0$

Key Insight

Comparing ROC curves allows evaluation of system sensitivity (d') independent of the response criterion (β).

Factors Affecting β and d'

Response Criterion (β)

Decision Bias & Strategy

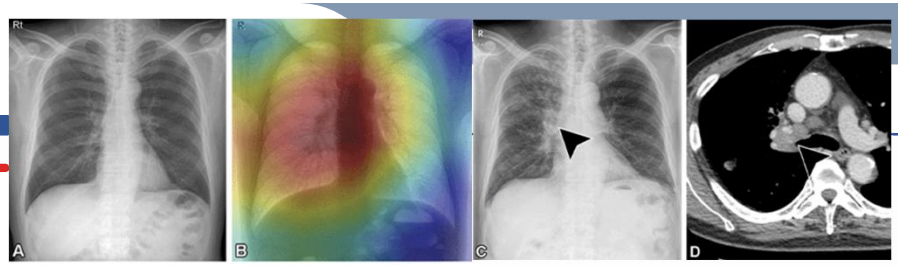
- **Signal Prevalence:** Rare signals lead to conservative bias (higher β), frequent signals lead to liberal bias.
- **Payoff Matrix:** Costs of Misses vs. False Alarms and benefits of Hits drive strategy shifts.
- **Instructions & Feedback:** Explicit directives or performance feedback can reset the criterion.
- **Organizational Policy:** Strict rules or safety culture influence risk-taking behavior.

Sensitivity (d')

Perceptual Ability & Discriminability

- **Signal Strength & Noise:** Physical intensity of signal versus background noise level (SNR).
- **Display Quality:** Resolution, contrast, and clarity of the information presentation.
- **Training & Experience:** Skilled operators have better internal templates for signal vs. noise.
- **Fatigue & Automation:** Time-on-task vigilance decrements and reliance on automated aids.

Case Study: Medical Imaging Diagnosis



The Challenge

In mammography or CT screening, the signal (tumor) is often faint and embedded in complex noise (dense tissue). The **cost of a "Miss" (undiagnosed cancer) is catastrophic**, while a **"False Alarm" (unnecessary biopsy) is costly but less severe**.

SDT Design Solutions

- **Adopt Liberal Criterion (Low β)**
  Radiologists intentionally set a risky bias to maximize Hits, accepting a higher rate of False Alarms to ensure no tumors are missed.
- **Increase Sensitivity (d')**
  Improve signal-to-noise ratio using higher resolution imaging, contrast agents, or Computer-Aided Detection (CAD) highlights.
- **False Alarm Management**
  Mitigate the cost of the liberal bias through "second reads" by other doctors or triage protocols before invasive procedures.

Payoff Logic

Screening Decision

Cost Analysis & Strategy

MISS (Type II)
Cost: Severe (Delayed treatment/Death)

FALSE ALARM (Type I)
Cost: Moderate (Anxiety/Biopsy)

STRATEGY
Set Liberal β (Bias towards "Yes")

Primary Goal
Minimize Misses at expense of False Alarms.

Case Study: Airport Security Screening

The Challenge: Low Prevalence Effect

Real threats are extremely rare events (low prevalence). When signals are rare, operators naturally shift their decision criterion (β) to become more conservative to avoid false alarms, paradoxically leading to increased Miss rates for actual threats.

Countermeasures

- **Threat Image Projection (TIP)**
  Artificially insert fictional threat images into baggage scans at random intervals to increase apparent signal frequency and keep vigilance high.
- **Performance Feedback**
  Provide immediate feedback on hits and misses during training and TIP events to calibrate the operator's criterion (β).
- **AI-Aided Pre-Screening**
  Use automated algorithms to highlight potential areas of interest, effectively boosting sensitivity (d') by cuing visual search.

Screening Performance

X-Ray Screening

Impact of Signal Probability

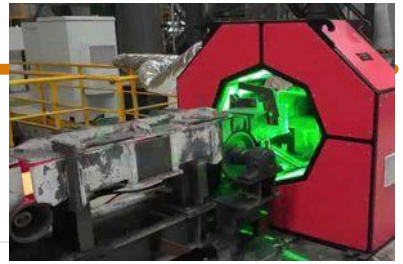
Low Prevalence
Conservative Criterion (High β) → More Misses

Artificially Increased
Balanced Criterion (Optimal β) → Better Detection

Design Goal
Combat vigilance decrement via artificial signals.

Application: Industrial Inspection & ATC

Industrial QC



Improve Contrast & Illumination

Enhancing physical signal strength increases **Sensitivity (d')** directly, making defects pop out from the background.

Standard Defects Library

Providing reference examples ("Limit Samples") helps inspectors maintain a consistent internal **Criterion (β)**.

Systematic Checklists

Reduces memory load and search uncertainty, ensuring all areas are scanned with equal attention.

Air Traffic Control

Alert Threshold Tuning

Adjusting software sensitivity to balance **Misses** (safety risk) vs. **False Alarms** (distraction) based on sector density.

Minimize Nuisance Alerts

Filtering out non-threats prevents "desensitization" where controllers ignore valid warnings due to high false alarm rates.

Multi-Sensor Fusion

Combining radar, ADS-B, and transponder data mathematically increases signal-to-noise ratio, effectively raising **d'**.

Improving Performance

Optimizing Detection Systems

1 Increase Sensitivity (d')

Enhance the distinguishability of the signal from noise.

- Maximize Signal-to-Noise Ratio (SNR)
- Improve training & feedback loops
- Increase dwell time for inspection
- Use redundant coding & better displays

✓ Better Signal Quality

2 Set Criterion (β)

Intentionally calibrate decision bias based on context.

- Align with Cost/Benefit matrix
- Adjust for signal probability
- Adapt thresholds dynamically by context

✓ Managed Bias

3 Support Decisions

Provide cognitive aids to reduce uncertainty and error.

- Implement ROC-based policies
- Use automated double-checks
- Design alarms with graded confidence

✓ Cognitive Aiding

Limitations & Considerations

Statistical Assumptions

Equal-Variance Assumption: The premise that Signal+Noise and Noise distributions have identical variance often fails in practice.

Non-Gaussian Distributions: Real-world data distributions are frequently skewed or heavy-tailed, unlike ideal normal curves.

Dynamic Environments

Nonstationary Environments: Signal probabilities and noise levels often fluctuate over time, unlike static lab conditions.

Learning Effects: Operators' sensitivity (d') and criteria (β) change as they gain experience or become fatigued.

Observation Dependence

Sequential Dependence: In many tasks, the probability of a signal at time *t* is related to time *t-1* (autocorrelation).

Decision History: Operators may be influenced by their recent sequence of decisions (e.g., reluctance to say "Yes" three times in a row).

Critical Warning

Lab to Field Generalization: Beware of overgeneralizing laboratory SDT results to complex field settings without validation.

Contextual factors (stress, workflow, interruptions) significantly alter performance parameters observed in controlled experiments.